

ELECTRIC VEHICLE DESIGN COURSE CONTENT

Duration- 4-6 Weeks

Centre of Excellence for Electric Vehicle Design - Course Curriculum

MODULE 1:

Introduction to Electric Vehicles

- History and evolution of EVs
- Types of electric vehicles (BEV, HEV, PHEV, FCEV)
- Overview of global and Indian EV ecosystem
- EV architecture and subsystems overview

MODULE 2:

Vehicle Dynamics and Chassis Design

- Basics of vehicle dynamics (acceleration, braking, cornering)
- Suspension, steering, and braking systems in EVs
- Lightweight materials and chassis design for EVs
- Structural integrity and crashworthiness

MODULE 3:

Electric Powertrain Design

- Electric motors used in EVs (BLDC, PMSM, Induction motors)
- Motor performance characteristics and selection
- Transmission systems for EVs (single/multi-speed)
- Regenerative braking and torque vectoring

MODULE 4:

Battery Technology and Management

- Battery types (Li-ion, LFP, NMC, solid-state, etc.)
- Battery pack design and integration
- Battery Management Systems (BMS): architecture, safety, thermal management
- Charging technologies (AC, DC, wireless, fast-charging protocols)

MODULE 5:

Power Electronics and Control Systems

- Inverters, converters, and controllers in EVs
- Vehicle Control Unit (VCU) and drive cycle control
- Motor control techniques (FOC, DTC, PWM)
- Thermal management of power electronics

MODULE 6:

Embedded Systems and Software for Evs

- Microcontrollers and embedded systems for EV control
- CAN communication protocol and diagnostics
- Real-time operating systems (RTOS) for EV applications
- Simulation tools (MATLAB/Simulink, LabVIEW)

MODULE 7:

Charging Infrastructure and Energy Management

- Charging station hardware and grid integration
- Smart charging and Vehicle-to-Grid (V2G) concepts
- Energy efficiency and management in EVs
- Policy, subsidies, and incentives for charging infra

MODULE 8:

Simulation, Modelling, and Prototyping

- CAD/CAE tools for EV design (SolidWorks, ANSYS, etc.)
- Simulation of motor and battery performance
- HIL (Hardware-in-the-Loop) and SIL testing
- Digital twin and real-time simulation of Evs

WEEK - 09

Standards, Regulations, and Safety

- ISO and IEC standards for EV design
- Functional safety (ISO 26262)
- Electrical, thermal, and crash safety in EVs
- EMI/EMC compliance for automotive electronics

WEEK -10

Capstone Projects and Industry Exposure

- Team-based projects on EV prototype or subsystems
- Collaboration with EV startups or manufacturers
- Industrial visits, guest lectures, and hands-on training
- Certification and portfolio development

Laboratory Facilities Required

- EV Powertrain Lab (motor, inverter, controller setup)
- Battery Testing Lab
- Vehicle Dynamics Simulation Lab
- Embedded System & CAN Bus Lab
- Prototyping & Fabrication Lab
- Charging Infrastructure Demonstration Setup